



Implementation Tools: Quick Copy

SAP Business One
Version 10.0

PUBLIC

THE BEST RUN 

In this topic, we will look at how to select and copy database records between SAP Business One company databases.

Objectives

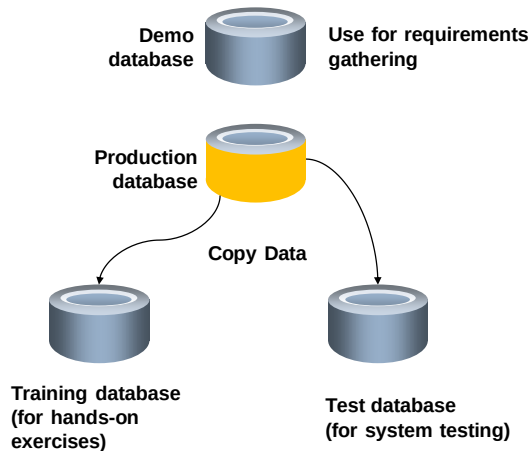


- At the end of this topic, you will be able to select and copy records between SAP Business One company databases using the Quick Copy tool

This topic covers the use of the Quick Copy tool during an implementation project. On completing this topic, you will be able to select and copy records between SAP Business One company databases.

Business Scenario

- During an implementation project you will typically use and create multiple company databases
- You will need separate database for testing and training purposes
- Quick Copy allows you to quickly copy subsets of data to other databases (individual records or categories of data)



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Quick Copy is a very useful tool during an implementation project. During the implementation project you will typically create multiple company databases. For example:

- A new production database that you will configure for the customer
- A copy of the production database for testing
- A copy of the production database for training

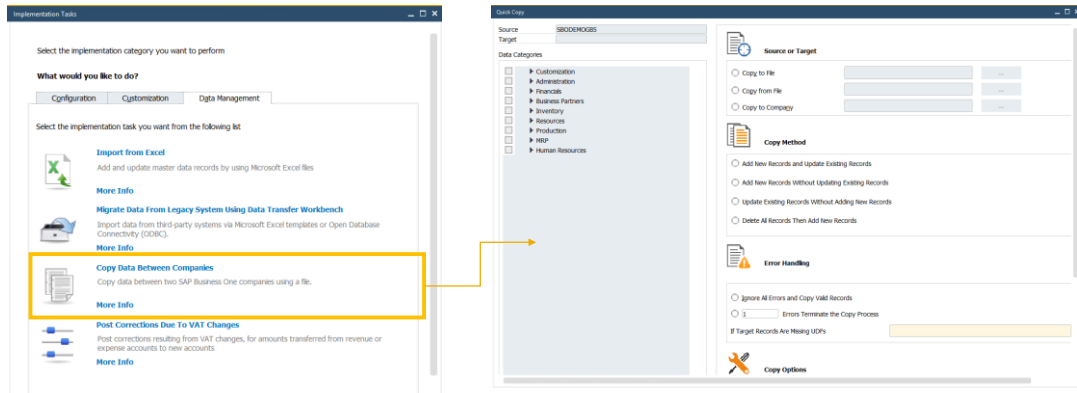
The Quick Copy tool enables you to quickly copy subsets of data from the production database to other databases. You can select individual records or entire categories of data.

For example, you might want to copy master data records and configuration settings from the production database to a test database. Or you can copy recent customizations you make on the production system to another database for testing and validation.

Note that Quick Copy is not a full database copy. You still need to maintain full database backups for recovery purposes.

Implementation Center

Administration > System Initialization > Implementation Center > Implementation Tasks > Data Management tab



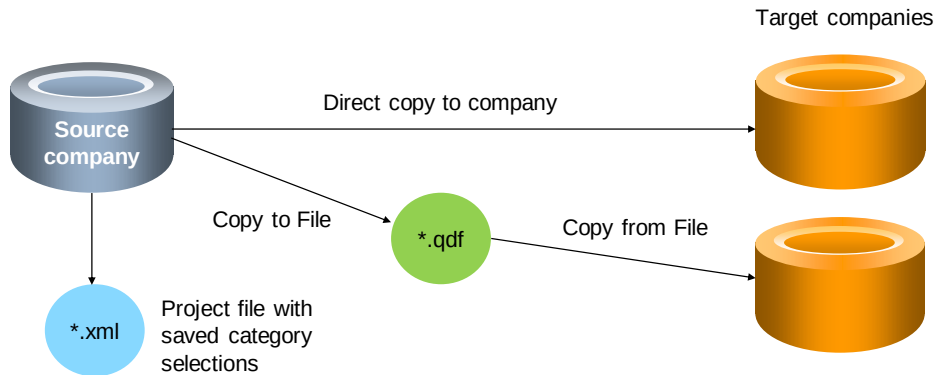
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Quick Copy is one of the tools provided in the Implementation Center in SAP Business One. To launch Quick Copy, choose *Copy Data Between Companies*.

Quick Copy Overview

- Direct copy from source to target database (same version required)
- Indirect copy (using an intermediate file)



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Quick Copy allows you to:

- Copy directly between company databases. The database version of the target company should match that of the source company.
- Copy indirectly via an intermediate file. If you choose indirect copy, Quick Copy saves the selected records in a file with the extension .qdf. You then login to the target company and run the import using Quick Copy.

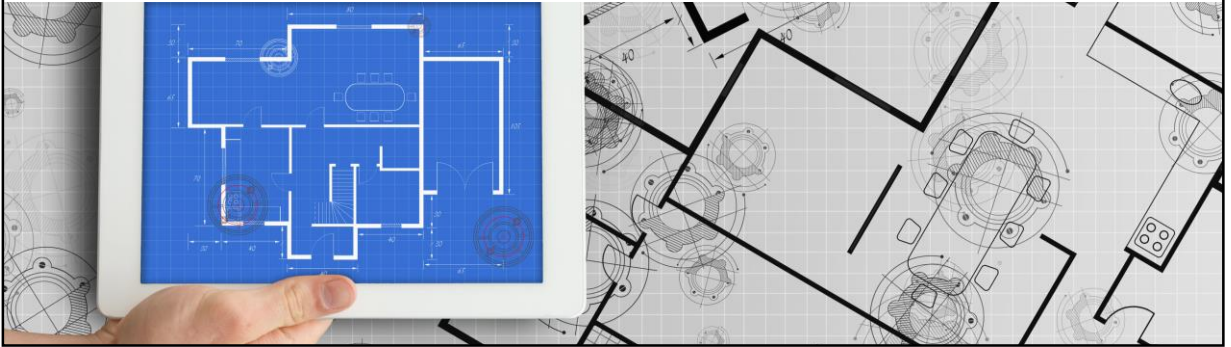
The company database that you are currently using is the source.

For direct copy, you will be prompted to select the target company on the database server and provide the login credentials.

For indirect copy, you will create the qdf file on the source, then login to the target company to import the qdf file.

You can optionally save an xml project file containing your data category selections and copy options. You can use this project file the next time you use Quick Copy, saving you from having to make the selections again.

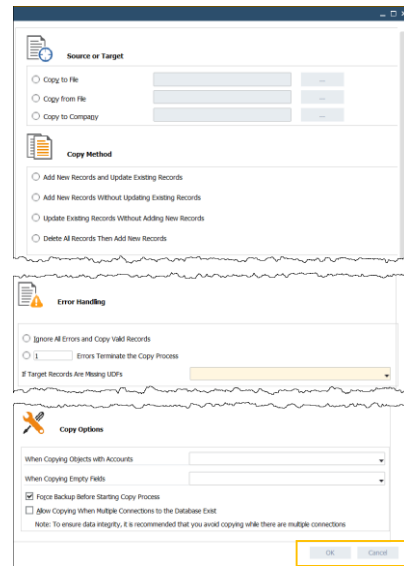
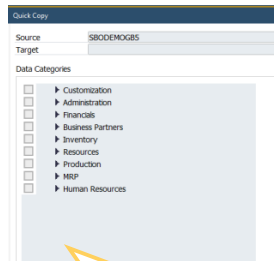
Copy Options



Quick Copy Window

Quick Copy window has two sections:

- Data categories are shown on the left
- Options for the copy are shown on the right
- Data categories are blocked from selection until you select all the required options and choose OK (the OK button is not active until the required options are set)



When you open the Quick Copy window you can see that the categories of data are shown on the left-hand side of the screen. The categories are initially blocked for selection until you select or set options on the right-hand side of the window.

After you make selections on the right side of the window, the OK button becomes active. Choose the OK button, and the data categories will open up for selection.

Options for Quick Copy

Source or Target	Select: ▪ Copy to File to copy records to an intermediate file (*.qdf) ▪ Copy from File to import records to a target company from a qdf file ▪ Copy to Company to copy directly
Copy Method	Defines how the import will insert or update records in the target database. Not applicable for Copy to File.
Error Handling	Defines how the system will handle errors encountered during either the copy or import process
Copy Options	Defines various options for the import. Not applicable for Copy to File.

The screenshot shows the 'Quick Copy' dialog box with the following sections:

- Source or Target:** Three radio buttons: 'Copy to File', 'Copy from File', and 'Copy to Company'. Each has a corresponding file selection button.
- Copy Method:** Four radio buttons: 'Add New Records and Update Existing Records', 'Add New Records Without Updating Existing Records', 'Update Existing Records Without Adding New Records', and 'Delete All Records Then Add New Records'.
- Error Handling:** Two radio buttons: 'Ignore All Errors and Copy Valid Records' and 'Errors Terminate the Copy Process'. Below is a dropdown menu for 'If Target Records Are Missing (QDF)'.
- Copy Options:** Two dropdown menus: 'When Copying Objects with Accounts' and 'When Copying Empty Fields'. Two checkboxes: 'Force Backup before Starting Copy Process' (checked) and 'Allow Copying When Multiple Connections to the Database Exist' (unchecked). A note at the bottom states: 'Note: To ensure data integrity, it is recommended that you avoid copying while there are multiple connections'. 'OK' and 'Cancel' buttons are at the bottom right.

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There are four sections with options:

- Select the copy function and browse to choose the qdf file or the target company database.
- Select the copy method (this applies only for Copy from File or Copy to Company)
- Set options for handling errors.
- Select the copy options (for import to the target company).

Once you have made these settings, and choose the OK button, the data categories on the left-hand side of the screen will open up for selection.

Copy Methods



Copy Method

- ☒ Add New Records and Update Existing Records
- ☐ Add New Records Without Updating Existing Records
- ☐ Update Existing Records Without Adding New Records
- ☐ Delete All Records Then Add New Records

Add new records and update existing records that have a matching key

And new records only but do not update records with a matching key

Only update existing records with a matching key

Erase all records for the categories before adding new records

The copy method works as follows:

- Add new records and update existing records that have a matching key
- Add new records but not update records with matching keys
- Only update existing records with matching keys without adding new records
- Erase all existing records for the categories in the target database before adding new records

Understanding Copy Methods

The effect of each copy method on the target database is shown here:

The source database T1 has four records (keys 1, 2, 3 and 4). The target database T2 has three records with matching keys 1,2 and 3.

Copy Method	Copy from Table 1 (T1) to Table 2 (T2)							
Add New Records and Update Existing Records	Before Copy				After Copy			
	T1		T2		T1		T2	
	Business Key	Content	Business Key	Content	Business Key	Content	Business Key	Content
	1	A	1	C	1	A	1	A
	2	B	2	F	2	B	2	B
Add New Records Without Updating Existing Records	Before Copy				After Copy			
	T1		T2		T1		T2	
	Business Key	Content	Business Key	Content	Business Key	Content	Business Key	Content
	1	A	1	A	1	A	1	A
	2	B	2	F	2	B	2	F
Update Existing Records Without Adding New Records	Before Copy				After Copy			
	T1		T2		T1		T2	
	Business Key	Content	Business Key	Content	Business Key	Content	Business Key	Content
	1	A	1	A	1	A	1	A
	2	B	2	F	2	B	2	F
Delete All Records Then Add New Records	Before Copy				After Copy			
	T1		T2		T1		T2	
	Business Key	Content	Business Key	Content	Business Key	Content	Business Key	Content
	1	A	1	A	1	A	1	A
	2	B	2	F	2	B	2	F

This slide shows the effect of each of the copy methods on the target company database:

- **Add New Records and Update Existing Records:** In the example, a new record is added to the target database T2 with key 4. The records with matching keys 1, 2 and 3 will be updated with the values from the source database T1. Notice that the content for record keys 1, 2 and 3 has changed.
- **Add New Records Without Updating Existing Records:** In the example, a new record is added to the target database T2 with key 4. However, unlike the first method shown above, in this case the records with keys 1, 2 and 3 are not updated.
- **Update Existing Records Without Adding New Records:** This method does not add any new records so record key 4 is not inserted in the target database. The records with matching keys 1, 2 and 3 will be updated with the values from the source database T1.
- **Delete All Records Then Add New Records:** This option deletes data for a matching key, then adds new records.

Error Handling



Error Handling

☐ Ignore All Errors and Copy Valid Records

☐ 1 Errors Terminate the Copy Process

If Target Records Are Missing UDFs

Do Not Copy Records with Missing UDFs

Do Not Copy Records with Missing UDFs

Copy Records and Ignore Missing UDFs

Error handling choices govern Quick Copy response while importing records:

- Ignore errors and continue copying valid records
- Terminate the copy process after a specified number of errors
- If the records contain UDFs, but the target system does not, you can elect to skip the records, or, copy them and ignore the missing UDFs

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The error handling choices govern the Quick Copy response to an error while importing the records. You can choose to:

- Skip the affected record and continue copying valid records, or
- Stop the copy process when the number of errors reaches a specified limit

If the objects to be copied contain user-defined fields (UDFs), and the target system does not contain the UDFs, you can elect to skip the records, or to copy the records and ignore the missing UDFs:

- If you select the Copy Records and Ignore Missing UDFs option, the application copies records even when user-defined fields in the source company are missing from the target company.
- If you select Do Not Copy Records with Missing UDFs, and for example, the master data in the source company contains a UDF that is not present in the target company, the copy operation will fail.

For example, the BP table in the source company contains a user-defined field, whose definition has not been copied to the target company. If this field is not selected, when you try to copy BP records to the target company, the copy operation fails.

Copy Options



Copy Options

When Copying Objects with Accounts		Use Accounts in Source Use Default Accounts in Target
When Copying Empty Fields		Do Not Overwrite Target Fields and Keep Original Values Overwrite Target Fields with Empty Values
☒ Force Backup Before Starting Copy Process ☐ Allow Copying When Multiple Connections to the Database Exist Note: To ensure data integrity, it is recommended that you avoid copying while there are multiple connections		

- If objects to be copied contain accounts, you can choose to copy the accounts to the target, or assign default accounts from the target
- If the objects contain empty fields, you can choose to skip copying the field and leave the original value in the target, or overwrite the target field with empty values
- Option to force backup before the import starts
- Option to copy to target even when there are active connections

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Many database records contain assigned accounts. The copy option for accounts can be chosen to:

- Use Accounts in Source – If the source accounts also exist in the target, then the application copies the account assignments from the object in the source to the object in the target. However, if the source accounts do not exist in the target, then the application encounters an error during the copy process.
- Use Default Accounts in Target – The application assigns the default accounts in the target to the object, and does not copy the source assignments. Note: If default accounts are not configured in the target company, and you select this option, then the account fields of the object in the target are empty after the copy process.

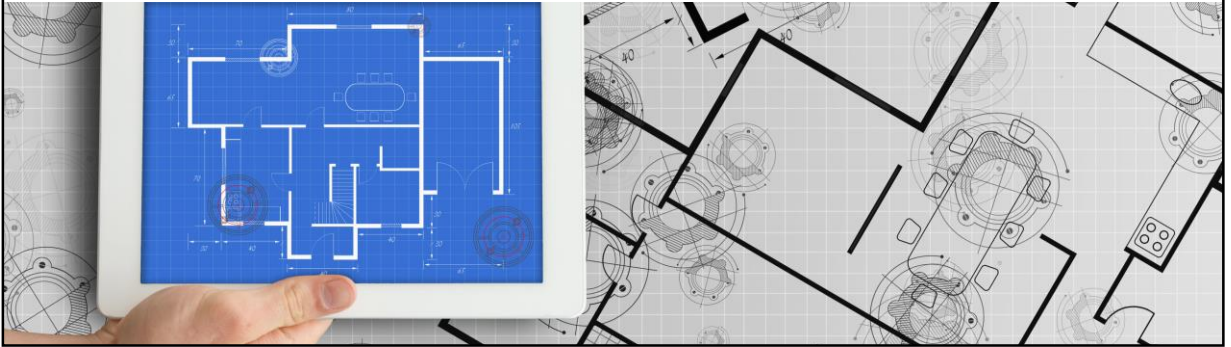
The copy option for empty fields only affects updates:

- Do Not Overwrite Target Fields and Keep Original Values – If a field in the source is empty, then the application skips copying this field. The application keeps the original value of the corresponding field in the target, which remains the same before and after the copy process.
- Overwrite Target Fields with Empty Values – If a field in the source is empty, then the application overwrites the target field with empty values.

You also have the option to:

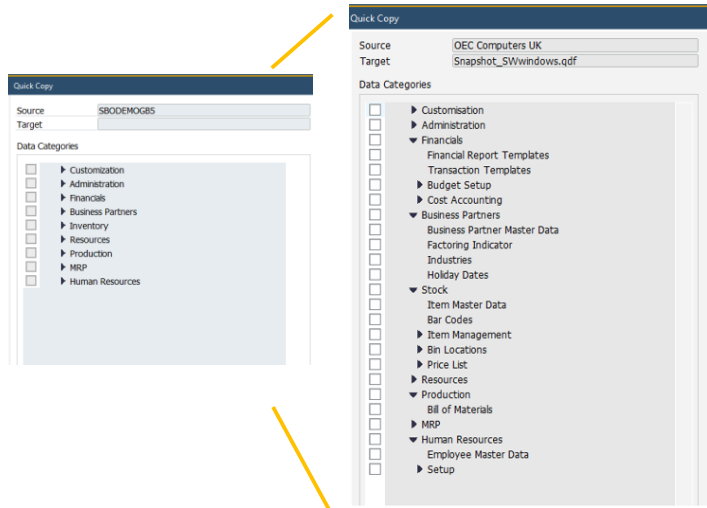
- Enforce a backup before the import starts. If you select the *Enforce Backup Before Starting Copy Process* checkbox, the import will not proceed if the target database has not been backed up within two hours.
- Allow the import of data to the target company database, even when there are user connections to it.

Categories and Dependencies



Category Structure

- Categories are shown in a tree structure that when expanded follows the SAP Business One menu structure



You select data from a category tree structure that when expanded follows the SAP Business One menu structure. Note that the structure does not change from company to company. The categories cover customizations (including queries and reports), administration and configuration settings, financial settings, business partner master data, inventory (including item master data and price lists), resource and production master data, and human resource master data.

Selecting Data for the Copy

- Select the categories required for copy
- The system shows data dependencies and you can choose to continue with the copy process, or adjust the selected categories

Object Name	Copy Method	No. of Records	Details
☒ File Formats	N/A	1	Record Details
☒ Languages	N/A	1	Record Details
☒ House Bank Accounts	N/A	2	Record Details
☒ Resource Master Data	N/A	9	Record Details
☒ Resource Groups	N/A	1	Record Details
☒ Route Stages	N/A	5	Record Details
☒ Units of Measurement	N/A	2	Record Details
☒ Units of Measurement	N/A	4	Record Details
☒ Bin Location Field Activ	N/A	2	Record Details
☒ Warehouse Sublevel C	N/A	3	Record Details
☒ Bin Location Master Da	N/A	2	Record Details

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You can select one or more categories at the top level of the category tree or at a lower level.

Choose the copy button to initiate the copy process.

The system first checks data dependencies and will show dependent objects that will be automatically copied together with the data you selected.

You can choose to continue with the copy process or cancel and adjust your selected categories.

Note: If the target database already contains records that reference G/L accounts, for example, business partner and item master data, journal entries posted by transactions, then you will not be able to import the chart of accounts from the source database

Viewing and Selecting Records to be Copied

- Before you run the copy you can view all records for a category
- You can deselect individual records from the copy process

Quick Copy						
☒	Customization	☒	A00003	J.B. Officeprint 1186	Items	J.B. Printers
☒	Query	☒	A00004	Rainbow ColorJet 5	Items	Rainbow Printers
☒	User-Defined Data	☒	A00005	Rainbow ColorJet 7.1	Items	Rainbow Printers
☒	Print Layouts and Reports	☒	A00006	Rainbow 1200 Laser	Items	Rainbow Printers
☒	Field Naming and Formatting	☒	E10000	Printer Label	Items	Items
☒	Multilanguage Translations	☒	C00001	Motherboard P4 Tur	Items	Items
☒	Cockpits and Dashboards	☒	C00002	Motherboard P4 Tur	Items	Items
☒	Administration	☒	C00003	Quadcore P4 2.4 Gh	Items	Items
☒	Financials	☒	C00004	Tower Case with Po	Items	Items
☒	Business Partners	☒	C00005	WLAN Card	Items	Items
☒	Inventory	☒	C00006	Network Card10/100	Items	Items
☒	Item Master Data	☒	C00007	Hard Disk Seagate 4k	Items	Storage
☒	Bar Codes	☒	C00008	Monitor 19" TFT	Items	Items
☒	Item Management	☒	C00009	Keyboard Comfort U	Items	Items
☒	Bin Locations	☒	C00010	Mouse USB	Items	Items
☒	Price List	☒	C00011	Memory DDR RAM 5	Items	Items
☒	Resources	☒	C00012	Cabinet for external	Items	Items
☒	Production	☒	C00013	1 TB disk for externa	Items	Items
☒	MRP	☒	C00014	Laptop case	Items	Accessories
☒	Human Resources	☒	D00001	Portable Hard Disk 1	Items	Storage
☒		☒	D00002	Portable Hard Disk 2	Items	Storage
☒		☒	E00001	Second hand Tablet	Items	PCs
☒		☒	I00001	DVD+R Disc 10-Pack	Items	Items
☒		☒	I00002	DVD+R Disc 50-Pack	Items	Items
☒		☒	I00003	USB Flashdrive 4GB	Items	Items
☒		☒	I00004	USB Flashdrive 8GB	Items	Items
☒		☒	I00005	J.B. Laptop Batteries	Items	Items
☒		☒	I00006	J.B. Laptop Batteries	Items	Items

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- To see the eligible records that you can copy, select the row for the object at the lowest level in a category. The list of records is shown on the right.
- You can individually deselect records that are not required for copy.

Dependent Categories



Inventory > Item Master Data



☐ Business Partner Master Data
BP Groups

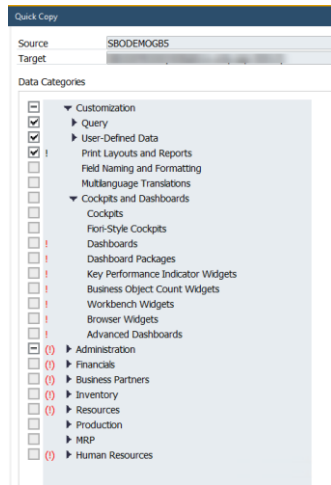
☐ Item Groups
Price Lists
Customs Groups
Manufacturers
etc.

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When you select a category, there might be data dependencies on other categories. For example, the item master data category has dependencies on business partner master data (vendors or resellers) as well as item groups and price lists.

Dependent Categories (Cont.)



- When you select an object or category, possible dependencies on other categories are marked
- Marked dependencies are derived from the database structure and are provided as a warning

! Selected category is dependent up on other categories

! Other categories are dependent on this category

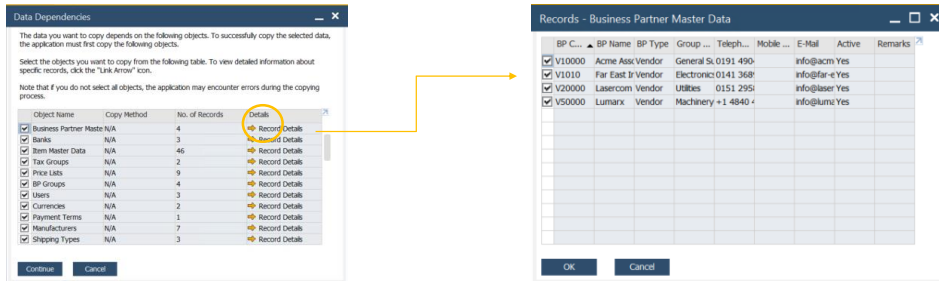
(!) Other categories are dependent on a category deeper in the structure

The Quick Copy tool indicates possible dependencies on other categories. These dependencies are derived from the database table structure, and not from the actual data records to be copied, and are therefore provided as warning for you to check if records in the marked category should also be copied. In the example shown, the item master data category is selected:

- A black exclamation mark '!' indicates that the item master data category is possibly dependent on other data categories.
- A red exclamation mark '!' indicates for an unselected category that the selected item master data category is possibly dependent on this category. In the example, the categories Bar Codes, Bin Location Master Data, and Price Lists are possibly dependent on the item master data category. Again this information is derived from the database table structure and may not represent the actual data.
- When you see a category marked with red parentheses '(!)', this indicates that the possible dependencies are deeper in the category structure. You can expand the category to see the dependent objects within its structure. In the example the Administration, Financials and Business Partners categories possibly have dependencies on the selected category, at a deeper level within the category.

Viewing Dependent Records

- During the copy process, the system displays actual dependent objects dynamically derived from the data records to be copied
- Dependent objects are automatically selected for copy with the other data
- Links are provided to view dependent records
- If you deselect a dependent object, copy may fail



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During the copy process, the system displays the real dependent objects, dynamically derived from the data records to be copied. These dependent objects are automatically selected for copy with the other data.

Links are provided to view the dependent records, enabling you to decide whether to include them in the copy.

Be aware that if you deselect a dependent object, the copy may fail.

Copy Log

- Log shows number of records copied, number of records failed, number of records not copied, and a link to drill down to records

Data Category	Object Name	Number of Records Copied	Number of Records Failed	Number of Records Not Copied	Details
Administration	Payment Terms	2	0	0	Result Details
Administration	Sales Employees/Buyers	1	0	0	Result Details
Administration	Units of Measurement	7	0	0	Result Details
Administration	Units of Measurement Groups	3	0	0	Result Details
Administration	User Groups	1	0	0	Result Details
Administration	Users	2	0	0	Result Details
Administration	Warehouse Sublevel Codes	2	3	0	Result Details
Administration	Warehouses	4	0	0	Result Details
Business Partners	Business Partner Master Data	5	0	0	Result Details
Customization	Flori-Style Cockpits	4	0	0	Result Details
Customization	Print Layouts and Reports	1	0	0	Result Details
Customization	Queries	8	0	0	Result Details
Customization	Query Authorization Groups	15	0	15	Result Details
Customization	Query Categories	3	1	3	Result Details
Customization	User-Defined Fields	8	0	0	Result Details
Customization	User-Defined Objects	2	0	0	Result Details
Customization	User-Defined Tables	5	0	0	Result Details
Inventory	Bin Codes	4	0	0	Result Details
Inventory	Bin Location Master Data	0	3	0	Result Details
Inventory	Item Master Data	70	0	0	Result Details
Inventory	Price Lists	6	0	0	Result Details

Record Key/Insta...	Result	Description	Help
S 3 L1	Copied failed	Warehouse sublevel is inactive; activate	
S 3 L2	Copied failed	Warehouse sublevel is inactive; activate	
S 1 A1	Copied successful		
S 1 A3	Copied successful		
S 2 S1	Copied failed	Warehouse sublevel is inactive; activate	

Record Key/Insta...	Result	Description	Help
05-A1-S1-L1	Copied failed	You cannot add or update the bin locat	
05-A1-S1-L2	Copied failed	You cannot add or update the bin locat	
05-A3-S1-L1	Copied failed	You cannot add or update the bin locat	
You cannot add or update the bin location; one or more warehouse sublevels are inactive			

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The copy process may take a long time if you are doing a direct copy and there are many records. For this reason it may be better to copy to an intermediate file so that you can perform the import at any time.

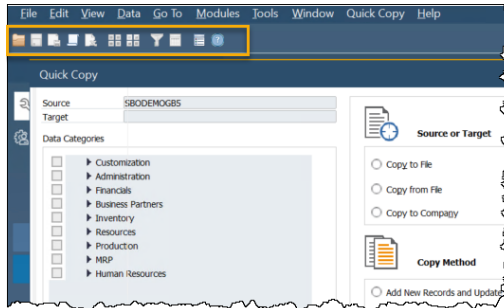
After the copy, the system provides a full log record of the number of records copied, the number of records that failed, and the number of records not copied, together with a link to the records.

Note: if you chose the option to stop after 1 error, you will only see records copies up to the point of failure.

In the example, shown, we can see the number of records copied for the selected Customization and Item Master Data categories, plus the dependent data categories. And we can drill-down to see the codes of the copied records.

Since the report shows an error on Bin Location master data, we can drill down to see the reason why this failed.

Quick Copy Window (cont.)



- Icon bar displays when Quick Copy window open
- Contains icons to view and display project files and log files
- Toggle dependencies on and off
- Access the copy options in the right-side of the window
- View online help

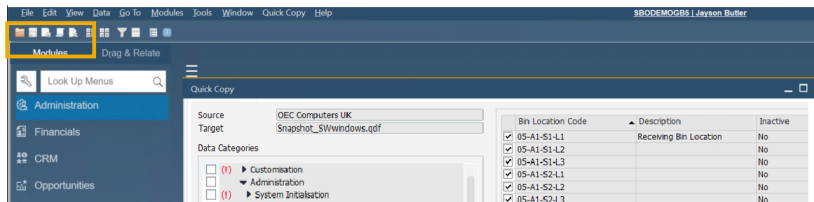
When the Quick Copy window is open you see a new icon bar in the client application.

The icons enable you to:

- Open a previously saved project XML file
- Save a new project file
- Toggle on and off the display of dependent categories
- Show the copy options in the right-side of the window. These options become hidden when the copy process has run, so you can use this icon to view them again.
- View the online help for Quick Copy

Saving a Project File

- Option to save settings and selected categories in a project file (XML)
- To reuse the project file, choose the toolbar icon



The XML project file contains the saved data categories and the selected copy options (copy methods, error handling, etc.). This file can be reused and can save time if you repeatedly need to export the same types of data.

The system will prompt you to create a project file if you close the Quick Copy window, or you can choose the save icon from the icon toolbar.

You can also create and save a project file when you import on the target company.

Summary



Here are some key points:

- Quick Copy allows you to selectively copy data records from one company database to another
- Quick Copy is useful during an implementation project for copying customizations and configuration information, and master data, to a test or training system or to a different company entity
- You can copy directly, or copy to an intermediate file:
 - For direct copy, both companies must have the same product version
 - The intermediate file has the extension qdf
- You can select one or more categories for copy. Note: Categories are not selectable until you complete all the required copy options on the right-side of the window.
- To see eligible records for copy, select the object row in the category hierarchy. You can deselect individual records.
- When you select a category for export, Quick Copy automatically marks dependent objects. During the copy process the dependent objects are also selected for copying.
- You can optionally save the selected categories and settings in an xml file, for repeat use.

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Here are some key points to take away from this session.

- Quick Copy allows you to selectively copy data records from one company database to another.
- The tool is useful during an implementation project for copying customizations and configuration information, or master data, to a test or training system, or to a different entity for the same company.
- You can copy directly, or copy to an intermediate file:
 - For direct copy, both companies must have the same product version
 - The intermediate file has the extension qdf
- You can select one or more categories for copy. **Note:** Categories are not selectable until you complete all the required copy options on the right-side of the window.
- To see eligible records for copy, select the object row in the category hierarchy. You can deselect individual records.
- When you select a category for copy, the Quick Copy tool automatically marks dependent objects. During the copy process the dependent objects are also selected for copying.
- You can optionally save an XML project file containing your selected data categories and copy options. This can save time if you repeatedly need to export the same types of data.

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